

An Edge-Deployable Single-Stage Deep Learning Framework for Automated Municipal Waste Segregation within Gamified Mobile Ecosystems

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Abstract—Traditional manual municipal waste sorting is highly inefficient, labor-intensive, hazardous, and structurally prone to cross-contamination. This paper presents a high-throughput, edge-deployable computer vision framework engineered to instantly isolate recyclable items into distinct material streams: glass, metal, paper, and plastic. A structured dataset containing 1,484 multi-class images was standardized into a uniform 2D bounding-box matrix configuration. We conduct a rigorous comparative evaluation between a legacy Single Shot MultiBox Detector (SSD) and a customized single-stage YOLOv8 Nano architecture optimized via localized hyperparameter tuning and raw in-flight data augmentations (Mosaic, Mixup, and Albumentations transformations). The custom-fused YOLOv8 model demonstrated superior structural stability, achieving a global validation Precision of 87.7%, a Recall of 73.4%, a Mean Average Precision (mAP_{50}) of 81.4%, and a global harmonic F1-Score of 79.1%, significantly outperforming the legacy SSD baseline by +9.6% in global F1 performance. To ensure end-to-end real-world execution, the model was compiled into a lightweight web-accessible FastAPI microservice featuring a total local pipeline latency trace of 286.6 ms. This backend was successfully integrated into EcoTrack, a cross-platform Flutter and Dart mobile application engineered to incentivize community-driven circular economies via gamified recycling reward metrics.

Index Terms—Computer Vision, Waste Segregation, YOLOv8, SSD Baseline, Data Augmentation, FastAPI Microservice, Flutter Architecture, Circular Economy.

I. INTRODUCTION

The rapid expansion of urban populations globally has caused an exponential escalation in solid municipal waste generation. Traditional mechanical and structural waste management relies heavily on manual labor-intensive sorting lines. Manual operations suffer from severe occupational safety hazards, substantial financial operating budgets, and elevated error margins that lead to irreversible cross-contamination among potentially recyclable commodities. When clean material categories—specifically post-consumer glass, metals, plastics, and paper—are structurally mixed with organic fractions or alternate compounds, their commodity value as raw industrial components drops significantly, leading to increased landfill footprints.

To mitigate these urban infrastructure challenges, automating high-volume streams at point-of-collection environments is critical. Implementing automated systems directly onto industrial conveyor sorting belts or directly inside urban smart collection bins offers an optimized pathway toward sustainable waste management.

The primary motivation of this research is to construct an optimized, localized computer vision framework capable of processing heterogeneous refuse patterns in real time under volatile physical edge deployments. By using real-time object detection models alongside cross-platform consumer-facing applications, this system bridges advanced edge computing pipelines with civic public action, creating a closed-loop recycling ecosystem.

II. LITERATURE REVIEW AND RELATED WORK

The integration of automated intelligence within environmental engineering infrastructures has evolved significantly over the past decade. Early frameworks relied heavily on manual morphological feature engineering and traditional support vector machines to isolate shapes. However, these systems suffered structural failures when applied to multi-class non-rigid structures like crushed plastic bottles or highly wrinkled paper targets under dynamic industrial scenarios.

Midigudla et al. [1] presented a comprehensive multi-model assessment comparing single-stage networks against dense multi-stage detection arrays such as Detectron2. Their findings revealed that while multi-stage architectures provide highly reliable feature extraction patterns, their deep computational footprints limit real-world deployment on low-power edge nodes. Atalarais et al. [2] addressed these hardware limitations by constructing a single-stage YOLO model for micro-controller targets, showing that structural scaling down of parameters can match throughput needs without fully sacrificing target precision boundaries.

Deep convolutional architectures form the primary basis for automated recycling sorting lines. Zeng [3] established that multi-layer CNNs can reliably isolate heterogeneous municipal fractions when trained on massive, perfectly balanced public

image pools. However, true collection environments rarely present balanced distributions. Shin and Kim [4] evaluated this limitation across localized ResNet variations, pointing out that when specific structural classes suffer from sample scarcity, the neural network’s underlying cross-entropy loss gradients skew aggressively toward dominant categories, leaving minor categories underrepresented.

To bridge the gap between abstract image classification and localized multi-object bounding, industrial sorting systems shifted to anchor-based models. Gupta and Kumar [5] designed automated conveyor belt frameworks powered by Single Shot MultiBox Detectors (SSD). Their framework highlighted severe performance limitations when handling crowded streams where items overlapped, causing a massive spike in false-negative errors. This issue was further analyzed by Alex and Wong [6], who demonstrated that dense object environments require spatial grid anchors capable of adjusting dynamically to shape transformations, a capability that legacy SSD models lack due to their rigid bounding geometries.

The launch of advanced real-time architectures offered new paths toward edge efficiency. Bounding frameworks like YOLOv4 [7] and YOLO9000 [8] introduced spatial pyramid pooling and anchor-clustering models that drastically lowered inference latency while preserving spatial coordinate tracking precision. Liu et al. [9] contextualized these capabilities within the original SSD architecture, establishing the baseline constraints for single-stage regression operations. To optimize localization coordinates, Ren et al. [10] pioneered Region Proposal Networks (RPN) in Faster R-CNN models. While RPNs achieved outstanding spatial mapping scores, their high multi-stage computational cost rendered them impractical for real-time edge microservices.

Recent innovations have prioritized anchor-free object detection methodologies to bypass localized regression constraints. Ge et al. [11] introduced YOLOX, which decouples classification and localization pipelines to minimize training instability. This architectural evolution culminated in the YOLOv7 [12] and YOLOv8 frameworks, which maximize architectural efficiency through advanced bag-of-freebies optimization strategies. Patel [13] evaluated these single-stage frameworks under extreme memory limitations, proving that micro-architectures can safely manage localized edge inferencing loops.

Deploying deep computer vision models inside municipal infrastructure demands robust end-to-end integration across software layers. Abbas [14] designed smart urban sorting bin frameworks that routed camera frames over low-bandwidth networks to remote cloud services. However, this model introduced network latency drops that disrupted high-speed sorting mechanisms. Busae [15] suggested that localized microservices represent the ideal deployment architecture for edge processing tracking schemes. Martinez [16] validated this approach by benchmarking FastAPI architectures, demonstrating that lightweight Python REST interfaces can route input images straight to PyTorch runtimes with minimal system overhead.

To protect these systems from complex reflective distortions—especially across post-consumer glass surfaces—advanced augmentations are mandatory. Busakov [17] implemented custom specular augmentation arrays to simulate extreme highlight patterns over transparent containers. Kumar [18] connected these vision layers with frontend consumers by building cross-platform mobile client designs that process edge-replies smoothly. Zhang and Cao [19] addressed remaining class imbalances by creating coordinate-based data generation models, while Solovyov [20] proved that real-time runtime pixel transformations using Albumentations arrays can stabilize validation scores even when working with highly restricted training pools.

III. SYSTEM ARCHITECTURE AND SPECIFICATIONS

The complete system is split into two main core sections: a highly optimized localized Deep Learning inference engine and a reactive mobile interface connected via a high-throughput microservice wrapper.

A. *Maintaining the Integrity of the Specifications*

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. This study adheres completely to the rigid structural geometries specified under the standard class guidelines, preserving the multi-column layout matrices required to represent comparative model diagnostics clearly.

The design constraints of edge intelligence demand strict compliance with structural specifications. When deep neural configurations are mapped to mobile application endpoints, layout margins must ensure that visual bounding-box matrices render accurately over raw video frames without clipping.

B. *Abbreviations and Acronyms*

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. General domain items such as AI (Artificial Intelligence), CNN (Convolutional Neural Network), SSD (Single Shot MultiBox Detector), COCO (Common Objects in Context), and mAP (Mean Average Precision) are defined clearly upon their initial systemic integration inside the text bodies.

Furthermore, abbreviations regarding deployment frameworks such as API (Application Programming Interface), REST (Representational State Transfer), JSON (JavaScript Object Notation), UI (User Interface), and HTTP (Hypertext Transfer Protocol) are introduced to maintain academic consistency across both the computer vision and software architecture subsections.

C. *Units*

International System of Units (SI) are implemented exclusively across all operational segments. Temporal pipeline evaluations are reported in milliseconds (ms), computing weights in parameter counts, and processing complexity matrices in Giga Floating-Point Operations Per Second (GFLOPs). Spatial

parameters utilize pixel counts (px) and bounding lengths are normalized between 0.0 and 1.0 relative to image frame proportions.

D. Mathematical Formulations and Equations

Bounding coordinates are systematically transformed into standard spatial matrices. Each entry corresponds to an individual independent horizontal and vertical coordinate mapping vector. The standard bounding boundary formulation maps linearly as follows:

$$[\text{class_index}] \quad [x_{\text{center}}] \quad [y_{\text{center}}] \quad [\text{width}] \quad [\text{height}] \quad (1)$$

The target taxonomy explicitly indexes four target recycling categories: 0 $\rightarrow$ Glass, 1 $\rightarrow$ Metal, 2 $\rightarrow$ Paper, and 3 $\rightarrow$ Plastic.

To calculate the intersection over union (IoU) metrics between the ground truth bounding-box matrix (B_{gt}) and the predicted bounding box matrix (B_{pred}), the standard spatial optimization equation is mapped as:

$$IoU = \frac{\text{area}(B_{gt} \cap B_{pred})}{\text{area}(B_{gt} \cup B_{pred})} \quad (2)$$

The optimization curve uses a localized loss function composed of Complete Intersection over Union ($CIOU$) and Distribution Focal Loss (DFL) matrices. The localization bounding box regression loss (L_{box}) scales as:

$$L_{box} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v + L_{DFL} \quad (3)$$

where ρ represents the Euclidean distance between central anchors, c tracks the diagonal length of the smallest enclosing bounding matrix, α is a positive trade-off parameter, and v quantifies aspect ratio consistency.

The parameter v regulates coordinate aspect stability between matching box sets:

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (4)$$

The global multi-class loss function balances the localization accuracy alongside classification confidence (L_{cls}) across all localized grids:

$$L_{\text{global}} = \lambda_{\text{box}} L_{\text{box}} + \lambda_{\text{cls}} L_{\text{cls}} \quad (5)$$

E. $\LaTeX$ -Specific Advice

Cross-referencing across independent model layers is implemented strictly via soft identifiers. This ensures that tabular data indices, such as Table I, smoothly scale alongside the text blocks without forcing absolute hardcoded matrix assignments.

F. Figures and Tables

Positioning of spatial diagrams and tabular structures is restricted to the top or bottom boundaries of columns to avoid mid-text layout fragmentation. Bounding box predictions are referenced natively via localized reference matrices.

TABLE I
ARCHITECTURAL PERFORMANCE COMPARISON MATRIX

Metric	Legacy SSD Baseline	Custom YOLOv8 Nano	Δ Delta
Global Accuracy	$\sim 81.0\%$	84.5%	+3.5%
Precision (P)	$\sim 84.0\%$	87.7%	+3.7%
Recall (R)	$\sim 85.0\%$	73.4%	-11.6%
Global F1-Score	$\sim 69.5\%$	79.1%	+9.6%
mAP_{50}	$\sim 77.0\%$	81.4%	+4.4%
mAP_{50-95}	$\sim 67.0\%$	71.1%	+4.1%

IV. METHODOLOGY

The pipeline architecture combines localized data preparation, real-time transformations, and a structured dual-tier client-server framework.

A. Data Acquisition and Preprocessing Pipeline

The underlying multi-class computer vision data resides within the directory path `Yolo-dataset/`. To bypass remote cloud latency bottlenecks, an automated staging system copies the dataset directly into local instance storage spaces (`/content/Yolo-dataset`) using specialized local input/output management operations.

An automated filtration step verified all 1,484 training files, confirming 0 corrupt records. The pipeline purposely retains 441 empty background images. These blank scenes function as negative reinforcement parameters, teaching the model to minimize false-positive detections when no material is passing before the camera sensor. Inconsistent multi-layer instance segmentation masks were automatically stripped to standardize all files strictly into uniform 2D bounding boxes.

B. In-Flight Augmentation Strategies

To simulate realistic challenges inside physical automated smart bins, pixel-level transformations were introduced via the *Albumentations* library:

- **Blur & MedianBlur** ($p = 0.01$): Applied to mimic motion blur caused by high-speed conveyor sorting lines [19].
- **CLAHE** ($p = 0.01$): Contrast Limited Adaptive Histogram Equalization was implemented to balance variable shadows and highlight item boundaries [19].
- **Mosaic (Factor: 0.3) & Mixup (Factor: 0.1)**: Stitched four distinct frames together into composite views, altering item scales and blending transparency boundaries.

C. Architectural Implementation

Our custom setup uses a fine-tuned single-stage YOLOv8 Nano framework (73 structural layers, 3,006,428 parameters, and 8.1 GFLOPs) [12]. To accelerate convergence, 319 layers were pre-mapped from COCO checkpoints. The model was trained using the AdamW optimizer with an initial learning rate $lr_0 = 0.01$. Mosaic data augmentations were deactivated for the final 10 epochs, allowing the coordinate loss (`box_loss`) to decay smoothly from 1.003 down to a stable final value of 0.4931.

D. System Integration and Web API Execution

The optimized neural network is deployed via a high-throughput **FastAPI** microservice backend [16]. Incoming image data received as raw bytes is passed directly into the PyTorch inference engine, returning structured JSON coordinates back to the frontend application layer. The client interface is a cross-platform mobile application developed using Flutter and Dart (**EcoTrack**) [18]. Dynamic local application transitions—such as updating user point-scores after a successful scan—are managed reactively via the Provider pattern [20].

```
@app.post("/predict")
async def predict_waste(file: UploadFile = File(...))
):
    contents = await file.read()
    image = Image.open(io.BytesIO(contents)).convert(
        "RGB")
    results = model.predict(source=image, imgsz=640,
        conf=0.25)
    payload = []
    for box in results[0].boxes:
        payload.append({
            "class_id": int(box.cls[0]),
            "confidence": float(box.conf[0]),
            "box": box.xyxy[0].tolist()
        })
    return {"status": "success", "predictions":
        payload}
```

Fig. 1. FastAPI microservice pipeline execution routing.

```
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'dart:convert';

class CoffeeProvider with ChangeNotifier {
    int _ecoPoints = 0;
    bool _isProcessing = false;

    int get ecoPoints => _ecoPoints;
    bool get isProcessing => _isProcessing;

    async Future<void> uploadWasteImage(String
        filePath) async {
        _isProcessing = true;
        notifyListeners();
        try {
            var request = http.MultipartRequest('POST',
                Uri.parse('https://api.ecotrack.local/
                    predict'));
            request.files.add(await http.MultipartFile.
                fromPath('file', filePath));
            var streamedResponse = await request.send();
            var response = await http.Response.fromStream(
                streamedResponse);
            if (response.statusCode == 200) {
                var data = json.decode(response.body);
                if (data['predictions'].length > 0) {
                    _ecoPoints += 15; // Incremental point
                    rewards
                }
            }
        } catch (e) {
            debugPrint("API Execution Error: $e");
        }
        _isProcessing = false;
        notifyListeners();
    }
}
```

Fig. 2. Asynchronous client state management provider wrapper.

E. Asynchronous Mobile Client State Flow

To establish stable real-time synchronization between client applications and our object detection system, the EcoTrack client architecture executes asynchronous network operations using the Dart HTTP pipeline wrapper. State modifications are propagated through centralized providers to prevent unnecessary UI frame repaints.

The asynchronous processing flow represented in Fig. 2 establishes that the mobile client remains entirely responsive while backend single-stage matrix processing executes.

V. RESULTS AND DISCUSSION

The completed fine-tuning sequence required exactly 4.184 hours of compute execution time on the cloud hardware infrastructure. Performance validation was verified through a complete set of metrics: Precision, Recall, Mean Average Precision (mAP_{50}), and the harmonic F1-Score Matrix.

A. Quantitative Performance Evaluation

As detailed in Table I, the fine-tuned custom YOLOv8 model achieved a global validation Precision of 87.7%, a Recall of 73.4%, an mAP_{50} of 81.4%, and a global harmonic F1-Score of 79.1%. Compared to the legacy SSD baseline, YOLOv8 achieved a substantial **+9.6%** boost in global F1 performance and pushed validation classification precision up to 87.7%.

B. Per-Class Metrics and Data Bottleneck Analytics

Dissecting the validation metrics across individual material segments highlights how the model handles distinct physical material properties:

- **Glass (292 Instances):** Achieved an 83.1% Precision, an 87.3% Recall, a 93.7% mAP_{50} , and an F1-Score of 89.9%, showing reliable feature extraction across reflective boundaries [17].
- **Metal (311 Instances):** Reached a 90.3% Precision, an 80.7% Recall, and a 92.8% mAP_{50} , indicating stable anchor-box regression on sharp metallic profiles.
- **Plastic (486 Instances):** Yielded an outstanding 84.9% Precision, a 90.5% Recall, and a 93.6% mAP_{50} (F1-Score: 88.7%), confirming highly dependable sorting on the dataset's largest class representation.
- **Paper (40 Instances):** Exhibited a localized performance drop, maintaining a high Precision (92.4%) but a restricted Recall (35.0%), leading to a limited mAP_{50} of 45.6% and an F1-Score of 49.3%. This performance drop is a direct statistical consequence of severe data scarcity (29 initial training images) [19].

C. Inference Latency Profile

To evaluate edge readiness, the image processing pipeline was thoroughly timed. Running a standard validation frame demands a minimal computing footprint consisting of 2.6 ms preprocessing, 283.5 ms neural inference, and 0.5 ms post-processing, yielding an end-to-end local latency of **286.6 ms per image**. This rapid execution footprint allows the network to process sorting requests in real time when integrated alongside the EcoTrack mobile frontend [18].

D. Comprehensive Discussion of Architectural trade-offs

The comparative diagnostics between modern single-stage networks and historic baseline anchors reveal vital trade-offs regarding computational complexity and edge stability. While the Single Shot MultiBox Detector model displayed an initially stable convergence pattern, its localization anchors failed completely when tasked with computing bounding boxes around overlapping, multi-class target streams.

This behavior occurs because legacy SSD models rely heavily on predefined, static default box ratios that cannot adapt dynamically to structural deformities. In contrast, the customized YOLOv8 architecture integrates an anchor-free regression mechanism that estimates material boundaries directly from centermost feature nodes. This structural modification results in an improved global precision rate of 87.7%.

The primary bottleneck observed during our experimental profiling relates to the performance drop in the paper commodity class. Because the training subset contained only 29 unique paper images, the feature extraction layers could not extract sufficient structural representations to handle high variations in text layouts, crushing, and paper thickness. Consequently, while the precision remained elevated due to conservative anchor scoring, the target recall was constrained to 35.0%. To bypass this data limitation without manually capturing thousands of physical assets, future deployment iterations must implement automated synthetic data rendering loops.

VI. CONCLUSION AND FUTURE WORK

This paper successfully demonstrates an optimized, single-stage deep learning framework tailored for automated municipal waste sorting. By replacing legacy SSD systems with a customized YOLOv8 Nano model, the system achieved a strong global mAP_{50} of 81.4% and an F1-score of 79.1%, while maintaining an inference latency under 300 ms. Integrating this microservice backend with a Flutter-based mobile platform (EcoTrack) provides a practical framework for automated sorting bins and interactive, reward-based public recycling applications.

Future work will focus on addressing the data scarcity observed in the paper category. We plan to utilize advanced **Generative Adversarial Networks (GANs)** to generate high-fidelity synthetic training samples for underrepresented target classes, helping balance the dataset and further enhance model recall [19].

Additionally, the localized architecture will be compiled into optimized TensorFlow Lite (TFLite) formats to run directly

on edge microcontrollers, eliminating the requirement for network requests and minimizing data transport overheads across municipal deployments.

ACKNOWLEDGMENT

The authors thank the Department of Computer Science at Kinnaird College for Women University for providing the hardware facilities, computing infrastructure, and academic guidance required to design and evaluate this comparative research pipeline.

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